

- 5 Mesophyll cells in plant leaves synthesise sucrose from the products of photosynthesis. The enzyme sucrose synthase catalyses the final reaction in the synthesis of sucrose.

(a) Name the **two** monosaccharides that are used to make sucrose.

..... [1]

- (b) Scientists studying a species of flowering plant investigated the movement of sucrose from mesophyll cells through phloem parenchyma cells and companion cells into sieve tubes.

Figure 5.1 shows the movement of sucrose from a mesophyll cell to a phloem sieve tube element in the vein of a leaf.

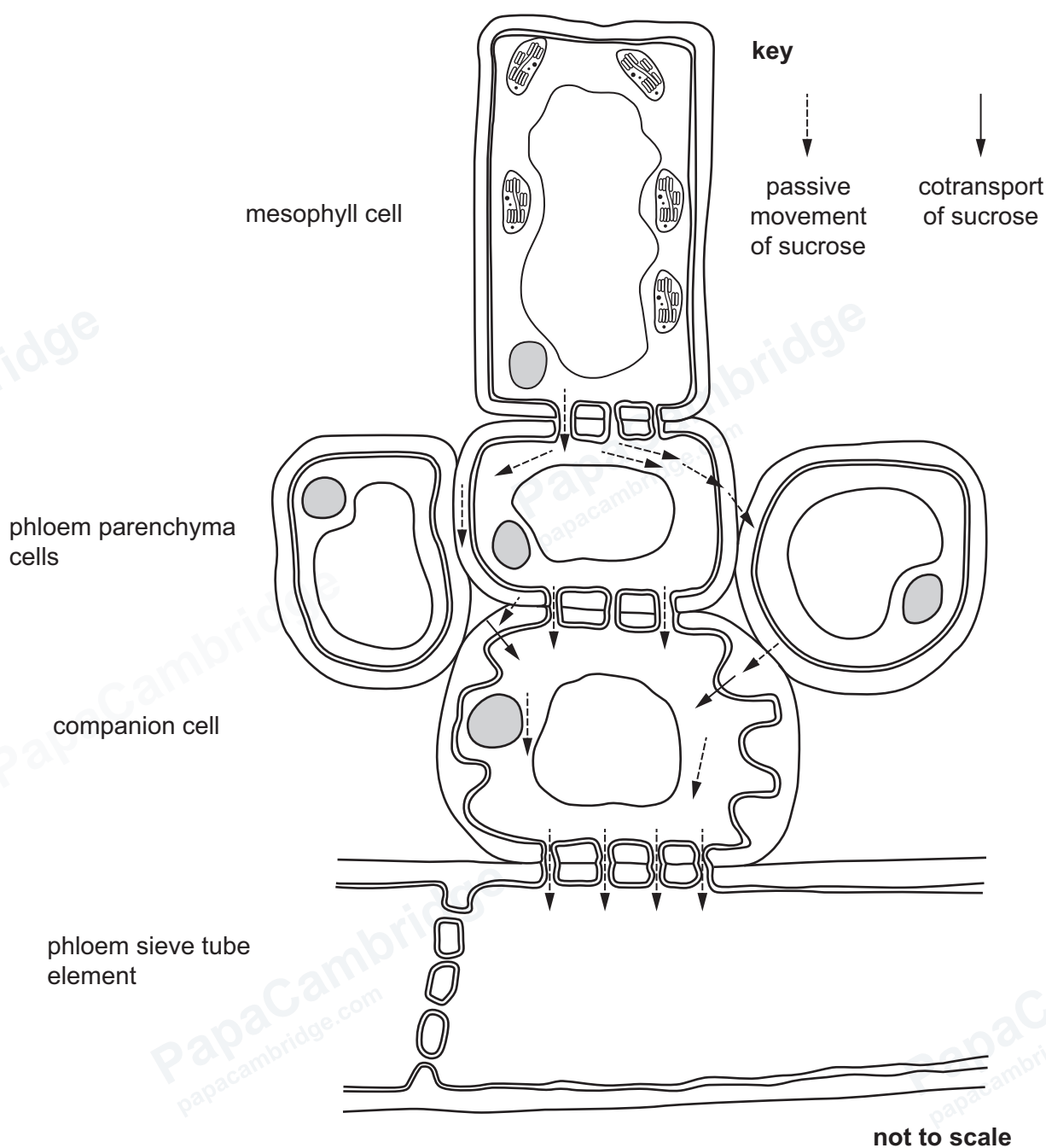
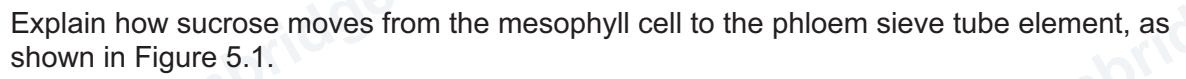


Figure 5.1



[5]

- (c) The scientists discovered a protein in the cell surface membrane of phloem parenchyma cells that transports sucrose out of the cells.

Figure 5.2 shows the structure of the membrane protein, known as SWEET, in the cell surface membrane.

Figure 5.3 shows the membrane protein viewed from the surface of the cell.

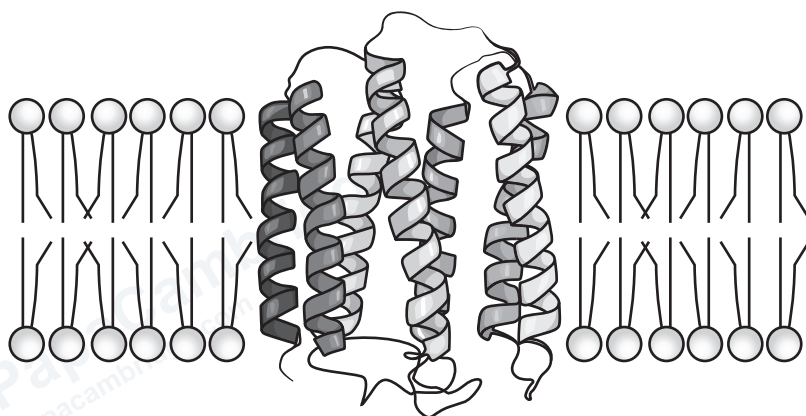


Figure 5.2

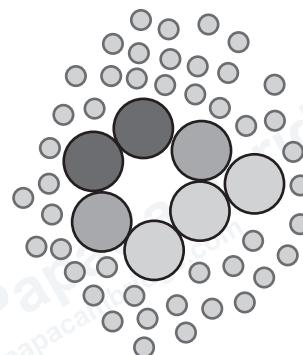


Figure 5.3

- (i) Describe the structure **and** arrangement of the SWEET protein as shown in Figure 5.2.

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..... [3]

- (ii) Explain why SWEET proteins are needed for the movement of sucrose out of the phloem parenchyma cell shown in Figure 5.1.

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..... [2]



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